A3 High Energy Physics Coursework Theory Report

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Below, "we" is used instead of "I" to follow the convention of scientific writing.

0. Introduction

Symmetries are central to modern physics. Lorentz invariance underpins special relativity, while local gauge symmetries such as SU(2) govern the structure of the electroweak interaction. When neural networks are used to represent physical fields, a generic architecture does not automatically respect these symmetries and can only learn them approximately from data. Following the coursework, this report studies two pre-trained networks, NN1 and NN2, that approximate an SU(2) scalar doublet and a gauge–boson field respectively. We first verify their transformation properties analytically and numerically (Sections 1–2), then construct alternative architectures in which equivariance is enforced exactly by restricting the network input to Lorentz invariants (Section 3). Finally, we build a classical Lagrangian from these equivariant fields and confirm its invariance under both Lorentz and SU(2) transformations (Section 4).

1. Q1: Equivariance of NN1: SU(2) Fundamental and Lorentz Scalar

NN1 is a generic fully connected network ($4 \to 128 \to 128 \to 128 \to 4$ with ReLU activations) that maps spacetime coordinates $x = (t, x, y, z) \in \mathbb{R}^4$ to four real outputs, which we package as a complex SU(2) doublet $\Phi = (\phi_1 + i\phi_2,\ \phi_3 + i\phi_4)^T \in \mathbb{C}^2$. Four real outputs map naturally to two complex components, matching the dimensionality of the fundamental representation of SU(2).

Lorentz scalar (analytical). A scalar field satisfies $\Phi(\Lambda x) = \Phi(x)$ for all Lorentz transformations $\Lambda \in \mathrm{SO}(3,1)$. This is a non-trivial constraint: a generic MLP takes all four components of x as independent inputs and could learn functions that are *not* Lorentz-invariant. For the scalar condition to hold, the network must effectively depend only on Lorentz-invariant input combinations, the simplest being $x_\mu x^\mu = t^2 - x^2 - y^2 - z^2$. Since NN1 has no architectural mechanism enforcing this (unlike the equivariant network in Section 3), the scalar property can only be approximate, arising from how the weights were tuned.

Lorentz scalar (numerical). We evaluate NN1 at 500 random points in $[-3, 3]^4$ and compute the relative error $\varepsilon = \|\Phi(\Lambda x) - \Phi(x)\|/\|\Phi(x)\|$ under 50 random transformations. Rotations are generated from a random axis (normalised Gaussian) and uniform angle in $[0, 2\pi]$; boosts use a random rapidity in $[0, 0.8]$ along a random direction. We restrict to transformed points within the training domain $|x_i| \leq 5$ and filter for $\|\Phi\| > 0.1$ to avoid inflated relative errors from near-zero outputs. The mean relative errors are approximately 5.6% for rotations and 4.2% for boosts, with maximum errors of approximately 6.6% in both cases, confirming that NN1 is an approximate Lorentz scalar.

SU(2) fundamental (analytical). The fundamental (spin-$\frac{1}{2}$) representation of SU(2) consists of 2×2 unitary matrices with unit determinant acting on vectors in $\mathbb{C}^2$. Under a global transformation $U \in \mathrm{SU}(2)$ the doublet transforms as $\Phi \to U\Phi$, which is the defining property of the fundamental representation. Crucially, SU(2) is an internal symmetry: it acts only on the field components (the output space) and not on the spacetime coordinates (the input), so the doublet structure is a consequence of how we package the four real outputs into $\mathbb{C}^2$ and places no constraint on the network weights. Unitarity ($U^\dagger U = I$) guarantees that $\Phi^\dagger\Phi$ is preserved, and the generators $\tau_a = \sigma_a/2$ (with σ_a the Pauli matrices) satisfy the SU(2) algebra $[\tau_a, \tau_b] = i\epsilon_{abc}\,\tau_c$.

SU(2) fundamental (numerical). Since SU(2) is internal, the transformation $\Phi \to U\Phi$ acts on the output and does not involve the network, so any equivariance test reduces to a check of the representation theory. We verify consistency by evaluating NN1 at sample points,

applying random $U \in \mathrm{SU}(2)$, and confirming $\Phi^\dagger\Phi = (U\Phi)^\dagger(U\Phi)$. Differences are of order 10^{-9} (Table 1), consistent with float32 precision. This test confirms unitarity of U rather than a learned property of the network, which is expected: for an internal symmetry the equivariance is *exact by construction* once the output is packaged as a doublet.

Table 1. SU(2) unitarity check for NN1: $|\Phi^\dagger\Phi - (U\Phi)^\dagger(U\Phi)|$ under random $U \in \mathrm{SU}(2)$.

Point	$\lvert\Phi\rvert^2$ before	$\lvert\Phi\rvert^2$ after U	$\lvert\Delta(\Phi^\dagger\Phi)\rvert$
0	1.028494	1.028494	9.01e-10
1	0.968938	0.968938	3.45e-08
2	0.071719	0.071719	1.66e-09
3	0.786423	0.786423	1.66e-09
4	0.815356	0.815356	1.73e-08

2. Q2: Equivariance of NN2: SU(2) Adjoint and Spin-1

NN2 is a generic MLP with the same architecture as NN1 but with 12 outputs, which we reshape into a 4×3 matrix $W_\mu^a(x)$ where $\mu = 0, 1, 2, 3$ is the Lorentz index and $a = 1, 2, 3$ is the SU(2) adjoint index (matching dim SU(2) = 3).

Lorentz spin-1 (analytical). A Lorentz vector (spin-1) field transforms as $W_\mu^a(\Lambda x) = \Lambda_\mu{}^\nu W_\nu^a(x)$: the 4×4 Lorentz matrix acts on the spacetime index μ while the internal index a is left untouched. As with NN1, the generic MLP architecture has no mechanism enforcing this equivariance, so it can only hold approximately.

Lorentz spin-1 (numerical). Using the same testing protocol as Section 1 (500 points, 50 transformations, $\|W\| > 0.1$ filter), we compute $\varepsilon = \|W_\mu^a(\Lambda x) - \Lambda_\mu{}^\nu W_\nu^a(x)\|/\|W(x)\|$. The mean relative errors are approximately 23% for rotations and 14% for boosts, with maximum errors of approximately 28% and 27% respectively, which are substantially larger than NN1 and indicates a coarser approximation. The error grows monotonically with the transformation angle (Figure 1). This is expected for a network trained to approximate equivariance: any smooth map can be expanded as $W_\mu^a(\Lambda x) = \Lambda_\mu{}^\nu W_\nu^a(x) + \delta W_\mu^a(x, \Lambda)$, where the violation δW vanishes at $\Lambda = I$ and grows with the size of the transformation. For a rotation by angle θ, $\Lambda = I + \theta\, K + \mathcal{O}(\theta^2)$ where K is the generator, so the leading violation is $\mathcal{O}(\theta)$, consistent with the approximately linear growth seen at small angles in Figure 1.

SU(2) adjoint (analytical). The adjoint representation of SU(2) acts on the three-dimensional space of the Lie algebra (spanned by the Pauli matrices σ^a). Under a global $U \in \mathrm{SU}(2)$ the representation matrix is a real 3×3 rotation $R^{ab} = \frac{1}{2}\,\mathrm{Tr}(\sigma^a\, U\, \sigma^b\, U^\dagger)$, and the field transforms as $W_\mu^a \to R^{ab}\, W_\mu^b$. As with Q1, SU(2) is internal and does not act on the coordinates, so this transformation law follows from identifying the index a with the Lie algebra basis. The matrix R is automatically a proper rotation ($\det R = 1$, $RR^T = I$), so the adjoint invariant $W_\mu^a\, W^{\mu\, a}$ is preserved.

SU(2) adjoint (numerical). As in Section 1, the SU(2) transformation is internal and acts only on the output, so the test reduces to verifying the representation theory. We confirm that the computed R matrices are proper rotations ($\det R = 1$, $\|RR^T - I\| \sim 10^{-15}$) and that the adjoint invariant $W_\mu^a\, W^{\mu\, a}$ is unchanged under random $R \in \mathrm{SO}(3)$. Differences are of order 10^{-8} (Table 2), consistent with machine precision and confirming that the adjoint structure is exact by construction once the index a is identified with the Lie algebra basis.

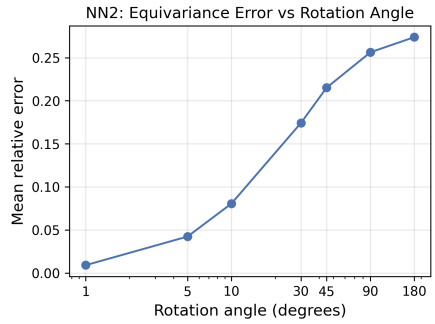

FIGURE 1. Relative error in $W_\mu^a(\Lambda x)$ vs transformation angle.

TABLE 2. SU(2) adjoint invariance check for NN2: $W_\mu^a W^{\mu a}$ before and after adjoint rotation R.

Point	$W_\mu^a W^{\mu\, a}$ before	after R	$\lvert\Delta\rvert$
0	0.037461	0.037461	9.03e-10
1	2.396670	2.396670	4.13e-07
2	0.407864	0.407864	1.52e-08
3	2.395378	2.395378	4.02e-08
4	0.848923	0.848923	4.31e-08

3. Q3: Construction and Training of Equivariant Networks

Rather than relying on weight tuning to learn symmetries approximately, we build them into the architecture so they hold exactly.

Analytical form.
We construct equivariant models of NN1 and NN2, denoted EquivariantPhi and EquivariantW, respectively.

EquivariantPhi. The only Lorentz-invariant scalar that can be formed from a single space-time point is $s = x_\mu x^\mu = t^2 - x^2 - y^2 - z^2$. We therefore define

$$\Phi(x) = \mathrm{MLP}(s) \in \mathbb{R}^4. \tag{1}$$

Since s is invariant under Lorentz transformations, $\Phi(\Lambda x) = \mathrm{MLP}(s(\Lambda x)) = \mathrm{MLP}(s(x)) = \Phi(x)$ by construction. The four real outputs are packaged as an SU(2) doublet exactly as in Section 1.

EquivariantW. The only Lorentz 4-vector available from a single point is x_μ itself. The most general equivariant ansatz is therefore

$$W_\mu^a(x) = f^a(s)\, x_\mu, \tag{2}$$

where $f^a(s)$ is an MLP mapping $s \to \mathbb{R}^3$ (three adjoint components). Under a Lorentz transformation, $W_\mu^a(\Lambda x) = f^a(s)\,(\Lambda x)_\mu = \Lambda_\mu{}^\nu\, f^a(s)\, x_\nu = \Lambda_\mu{}^\nu\, W_\nu^a(x)$, which is the correct spin-1 transformation law, holding exactly by construction. The SU(2) adjoint structure is inherited from the index a as in Section 2.

Implementation and results.
Both MLPs use three hidden layers of 128 units with ReLU activations. We generate training data by evaluating NN1/NN2 on 30,000 random points in $[-5, 5]^4$ (5,000 for validation) and train with Adam (lr $= 10^{-3}$) and ReduceLROnPlateau for 150 epochs (Figure 2).

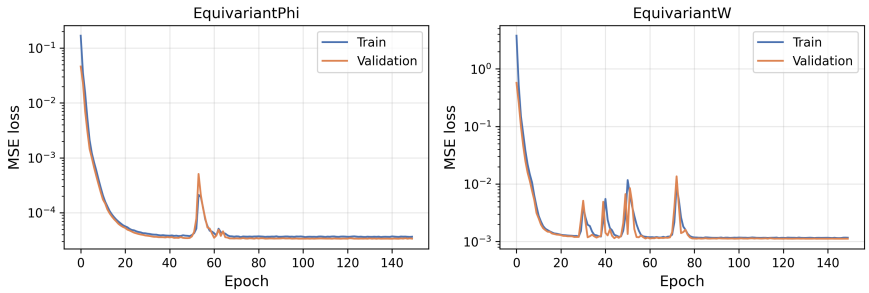

FIGURE 2. Training curves for EquivariantPhi and EquivariantW.

Verification of exact equivariance. Repeating the tests from Sections 1–2, the Lorentz equivariance errors are now of order 10^{-7} (Table 3), consistent with float32 precision and representing an improvement of five to six orders of magnitude over the original networks.

TABLE 3. Lorentz equivariance errors: original vs equivariant architectures.

Model	Transformation	Original	Equivariant
Φ	SO(3) rotation	0.0562	1.99e-07
Φ	Lorentz boost	0.0416	2.60e-07
W	SO(3) rotation	0.2302	3.45e-07
W	Lorentz boost	0.1415	4.80e-07

Emulation quality. On the validation set, EquivariantPhi achieves a mean relative error of 5.3% (median 3.5%), while EquivariantW gives 30.9% (median 15.7%). The higher error for W is expected: Equation (2) is the most general exactly equivariant form, but NN2 is only approximately equivariant, so its non-equivariant components cannot be captured and a residual error is unavoidable.

4. Q4: Invariant Lagrangian Construction

Analytic form.
We construct a classical Lagrangian with kinetic and mass terms for both fields:

$$\mathcal{L} = \partial_\mu \Phi^\dagger \partial^\mu \Phi - m_\Phi^2\, \Phi^\dagger \Phi - \tfrac{1}{4}\, G^a_{\mu\nu}\, G^{\mu\nu\, a} + \tfrac{1}{2}\, m_W^2\, W^a_\mu\, W^{\mu\, a}, \tag{3}$$

where $G^a_{\mu\nu} = \partial_\mu W^a_\nu - \partial_\nu W^a_\mu$ is the Abelian field-strength tensor. We omit the non-Abelian $g\epsilon^{abc} W^b_\mu W^c_\nu$ self-interaction since we work with a global (not gauged) SU(2) symmetry and the Abelian kinetic term already demonstrates the required invariance structure.

Lorentz invariance. Every Lorentz index is contracted: $\partial_\mu \Phi^\dagger \partial^\mu \Phi$ contracts two vector indices via $\eta^{\mu\nu}$; $G^a_{\mu\nu} G^{\mu\nu\, a}$ contracts two antisymmetric tensor pairs; and $W^a_\mu W^{\mu\, a}$ contracts one pair. Since Φ is a scalar and W^a_μ a vector, each term is a Lorentz scalar.

SU(2) invariance. $\Phi^\dagger \Phi$ is invariant under $\Phi \to U\Phi$ by unitarity. All adjoint indices a are contracted with δ^{ab}, preserved under $W^a \to R^{ab} W^b$ by orthogonality of R.

Implementation and results.
The equivariant networks output W_a^μ with an upper Lorentz index; we lower with $W_\nu^a = \eta_{\nu\mu} W_a^\mu$ before forming $G_{\mu\nu}^a$. All derivatives are computed exactly via PyTorch autograd. To ensure well-defined second derivatives we retrain with SiLU activations in place of ReLU, achieving comparable emulation quality (Figure 3). We set $m_\Phi = m_W = 1$.

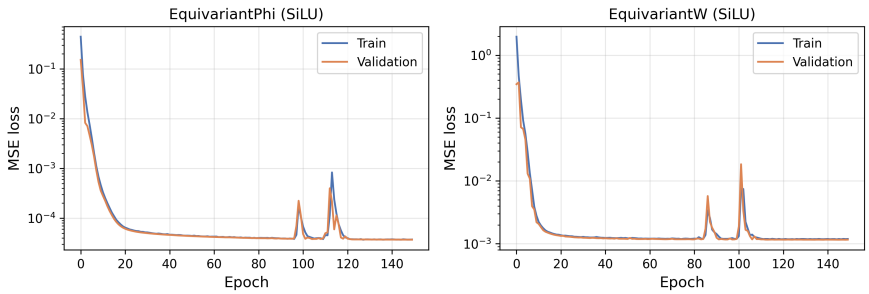

FIGURE 3. Training curves for SiLU equivariant networks.

Lorentz invariance (numerical). We evaluate $\mathcal{L}(x)$ and $\mathcal{L}(\Lambda x)$ at 15 random points under random SO(3) rotations and Lorentz boosts. The mean relative errors are 1.7×10^{-7} and 2.7×10^{-7} respectively, confirming that the Lagrangian is a Lorentz scalar to machine precision.

SU(2) invariance (numerical). Under random $U \in \mathrm{SU}(2)$, the maximum deviations in $\Phi^\dagger\Phi$ and $W_\mu^a W^{\mu\, a}$ are 4.7×10^{-8} and 2.3×10^{-7} respectively, consistent with float32 precision. This follows because the Lagrangian depends on the fields only through unitarily invariant ($\Phi^\dagger\Phi$) and orthogonally invariant (δ_{ab}) contractions.

USE OF GENERATIVE AI

Claude (Anthropic) was used to assist with code development and report writing. All outputs were reviewed and tested by the author, who takes full responsibility for the final submission.